Shubham Sah

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Education

Indian Institute of Information Technology (IIIT) — **Bhopal, India**
Bachelor of Technology in Computer Science and Engineering **CGPA: 7.59** — *Aug. 2023 – May 2027*

Experience

Blesso — **Bengaluru, India**
Software Development Engineer (SDE) Intern — *Jul. 2026 – Present*

- **Architected** an AEO/GEO visibility engine tracking brand citation rank across **ChatGPT, Perplexity, Gemini, Claude**, pairing deterministic entity resolution with selective LLM disambiguation to keep a 50-prompt scan under $9.
- **Developed** an automated Google Calendar sync service consuming change webhooks, using email/domain-fallback matching and upsert logic to attribute meeting records to the right account prospect without creating duplicates.
- **Engineered** a programmatic SEO engine using LLM-driven research and custom JSON schemas to mass-produce **1,000+** SEO/GEO/AEO-optimized blogs across collections via reusable rendering templates.

AlgoUniversity — **Hyderabad, India**
Software Engineer (SWE) Intern — *May 2026 – Jul. 2026*

- **Curated** a target list of **200+** high-potential local businesses via automated web scraping and market research.
- **Developed** 5+ responsive, reusable React landing page templates, cutting onboarding-to-deployment time by **45%**.

Indian Institute of Information Technology (IIIT) — **Bhopal, India**
Undergraduate Researcher (Quantum-Safe ZKP) — *Aug. 2025 – May 2026*

- **Designed** a lattice-based Zero-Knowledge Proof (ZKP) protocol prototype in Python to evaluate post-quantum schemes.
- **Orchestrated** a hybrid blockchain storage architecture pairing off-chain data vectors with on-chain Merkle roots, improving data retrieval speed by **25%** while cutting per-transaction storage and verification cost.

Projects

Recoup — AI Payment Recovery Agent | ***TypeScript, Bun, React, Razorpay API*** — GitHub | Live

- **Engineered** an autonomous payment-recovery agent with **exactly-once external execution**: SHA-256 action keys, atomic pre-call claims, and reference-ID reconciliation, so an ambiguous provider timeout never duplicates a charge.
- **Designed** a counterfactually-valid A/B evaluation over 2,000 simulated cases, measuring **+75.8% net value (+INR 32.4L)** against a fixed-retry baseline while issuing **65% fewer** customer actions.
- **Shipped** a deterministic safety guard (fraud freeze, opt-out, contact caps, quiet hours, expected-value floor) and a React/Three.js audit dashboard, backed by **98 automated tests** across a 9,300-line TypeScript codebase.

Remote Browser Control | ***React, Node.js, Docker, WebSockets, Puppeteer, AWS*** — GitHub | Live

- **Architected** a real-time remote browser execution pipeline using Node.js, WebSockets, and Puppeteer to stream containerized, high-performance Chromium instances to an HTML5 canvas in any browser.
- **Optimized latency to sub-100ms** by designing client-side mouse coordinate normalization and keyboard emulation.
- **Reduced server resource overhead by 30%** by debugging WebSocket connection leaks and enforcing automated Docker container cleanup tied to session lifespan, preventing orphaned instances.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash, HTML/CSS
Backend & Security: Django, Node.js, REST APIs, Webhooks (HMAC-SHA256), Zero-Knowledge Proofs, Idempotency
Infrastructure & DevOps: AWS, GCP, Docker, CI/CD (GitHub Actions), Git/GitHub, Linux, Bun
AI & Data: LLM Integration, Agentic Workflows, A/B Evaluation, SQLite/MySQL, MongoDB, Web Scraping
Frontend & Tools: React, Three.js, VS Code, Postman, Unit Testing, Agile/Scrum, LaTeX
Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Applied Cryptography

Publications

- **Quantum-Safe ZKP Architecture for Secure Verification** — Co-authored research on a lattice-based Zero-Knowledge Proof protocol for post-quantum verification; submitted to ***Cluster Computing Journal***.
- **Arduino-Based Robotic Car** — Embedded control with ultrasonic obstacle avoidance; ***IEEE SCEECS 2024***.

Achievements

- **Ranked** in the top 15 of 500+ students in competitive coding at IIIT Bhopal; **solved 770+ algorithmic problems** across LeetCode, Codeforces (Peak: 1863), and CodeChef (5-Star / Max 2000+).
- **Contributed** open-source codebase updates to distributed repositories during GirlScript Summer of Code (GSSoC).